

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Enterprise Java (EJ)

Practical – IX

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Class: TYIT	Batch: 1
Date of Assignment: 08/08/2026	Date/Time of Submission: 08/08/2026

9. Implement the following JPA applications

a. Develop a Simple Inventory Application Using JPA

Steps:

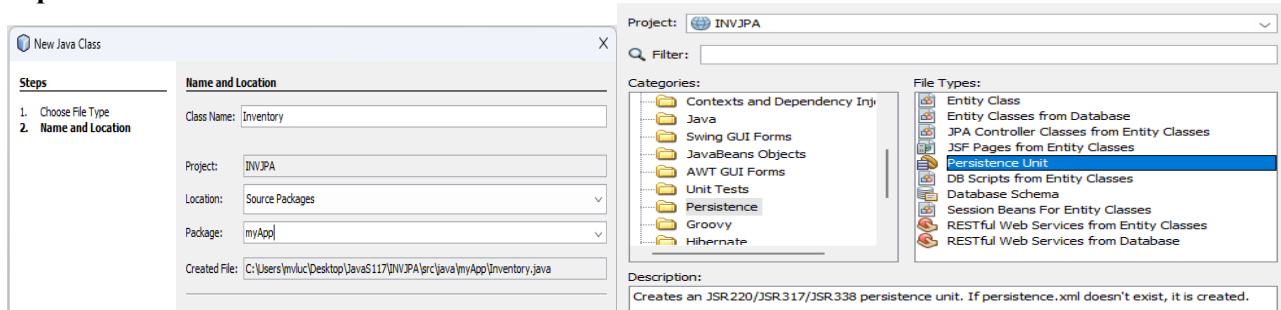
1. Create a Java Web Project named INVJPA.
2. Create Inventory database and inventory table in MySQL.
3. Add MySQL Connector and configure the JPA Persistence Unit.
4. Create Inventory.java Entity class.
5. Create index.jsp for entering item details.
6. Create checkInventory.jsp to save and display inventory records.
7. Run the project on Tomcat/GlassFish and test the application.

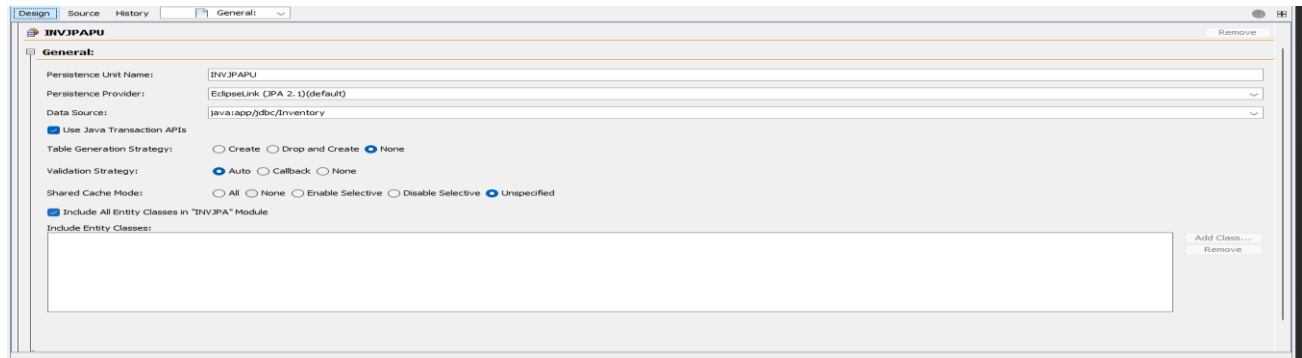
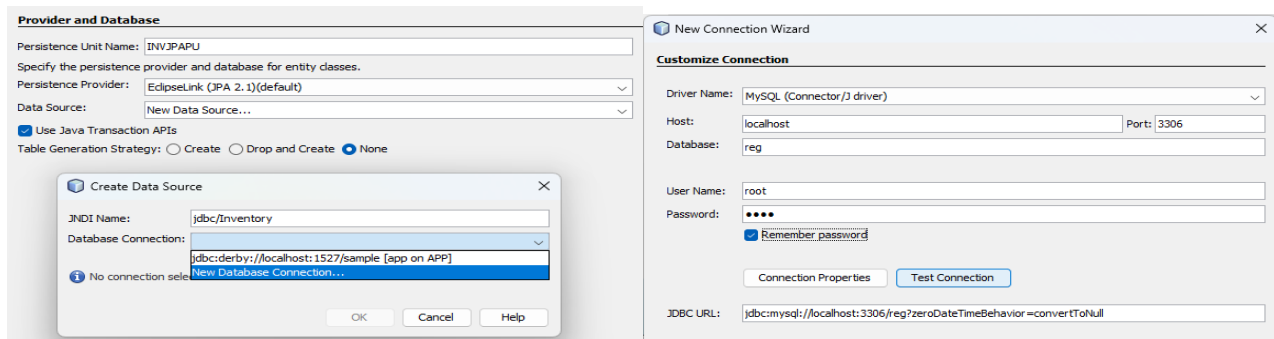
Code:

1. Inventory.java

```
package myApp;
import javax.persistence.*;
@Entity
@Table(name="Inventory")
public class Inventory {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    @Column(name="itemid", unique=true, updatable=false)
    private Integer itemid;
    @Column(name="itemName")
    private String itemName;
    @Column(name="description")
    private String description;
    @Column(name="quantity")
    private Integer quantity;
    @Column(name="addedDate")
    private String addedDate;
    // Constructor
    public Inventory() {}
    // Getters & Setters
    public Integer getItemID() { return itemid; }
    public void setItemID(Integer itemid) { this.itemid = itemid; }
    public String getItemName() { return itemName; }
    public void setItemName(String itemName) { this.itemName = itemName; }
    public String getDescription() { return description; }
    public void setDescription(String description) { this.description = description; }
    public Integer getQuantity() { return quantity; }
    public void setQuantity(Integer quantity) { this.quantity = quantity; }
    public String getAddedDate() { return addedDate; }
    public void setAddedDate(String addedDate) { this.addedDate = addedDate; }
}
```

2. persistence.xml





```
<?xml version="1.0" encoding="UTF-8"?>
<persistence version="2.1"
  xmlns="http://xmlns.jcp.org/xml/ns/persistence"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/persistence
http://xmlns.jcp.org/xml/ns/persistence/persistence_2_1.xsd">

  <persistence-unit name="INVJPAPU" transaction-type="RESOURCE_LOCAL">
    <non-jta-data-source>jdbc/Inventory</non-jta-data-source>
    <exclude-unlisted-classes>false</exclude-unlisted-classes>
    <properties/>
  </persistence-unit>
</persistence>
```

3. index.jsp

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<title>Inventory</title>
</head>
<body>
<form action="checkInventory.jsp" method="post">
<table>
<tr><td colspan="2" align="center"><b>Add the item in Inventory</b></td></tr>
<tr><td>Item Name:</td>
<td><input name="itemname" maxlength="25" size="50" /></td></tr>
<tr><td>Description:</td>
<td><textarea rows="5" cols="36" name="description"></textarea></td></tr>
<tr><td>Quantity:</td>
<td><input name="quantity" maxlength="4" size="5" /></td></tr>
<tr><td colspan="2">
<input type="submit" name="AddToInventory" value="AddToInventory" />
</td></tr>
</table>
</form>
</body>
</html>
```

4. checkInventory.jsp

```

<%@page import="java.util.*,javax.persistence.*,myApp.Inventory"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<DOCTYPE html>
<%!
private EntityManagerFactory eMF;
private EntityManager eM;
private EntityTransaction eT;
List<Inventory> inv;
%>
<%
eMF = Persistence.createEntityManagerFactory("INVJPAPU");
eM = eMF.createEntityManager();
String submit = request.getParameter("AddToInventory");
if(submit != null && ("AddToInventory").equals(submit)) {
    try {
        String itemname = request.getParameter("itemname");
        String description = request.getParameter("description");
        int quantity = Integer.parseInt(request.getParameter("quantity"));
        String addedDate = new java.util.Date().toString();
        Inventory invObj = new Inventory();
        invObj.setItemName(itemname);
        invObj.setDescription(description);
        invObj.setQuantity(quantity);
        invObj.setAddedDate(addedDate);
        eT = eM.getTransaction();
        eT.begin();
        eM.persist(invObj);
        eT.commit();
    } catch (RuntimeException e) {
        if(eT != null) eT.rollback();
        throw e;
    }
}
response.sendRedirect("checkInventory.jsp");
try {
    inv = eM.createQuery("SELECT i FROM Inventory i").getResultList();
} catch (RuntimeException e) {
    throw e;
}
eM.close();
%>
<html>
<head><title>Inventory</title></head>
<body>
<table border="1">
<tr><td colspan="5" align="center">
<b>Click <a href="index.jsp">here</a> to add items in the Inventory.</b>
</td></tr>
<tr><td>Item No</td><td>Item Name</td><td>Description</td><td>Quantity</td><td>Added Date & Time</td></tr>
<%
Iterator iterator = inv.iterator();
while (iterator.hasNext()) {
    Inventory objInv = (Inventory) iterator.next();
%>
<tr>
<td><b><%= objInv.getItemID() %></b></td>
<td><b><%= objInv.getItemName() %></b></td>
<td><b><%= objInv.getDescription() %></b></td>
<td><b><%= objInv.getQuantity() %></b></td>
<td><b><%= objInv.getAddedDate() %></b></td>
</tr>
<%
}
%>
</table>
</body>
</html>

```

5.MySQL :

```

mysql> CREATE DATABASE Inventory;
Query OK, 1 row affected (0.00 sec)

mysql> USE Inventory;
Database changed
mysql> CREATE TABLE inventory (
-> itemid INT PRIMARY KEY AUTO_INCREMENT,
-> itemName VARCHAR(50),
-> description VARCHAR(200),
-> quantity INT(40),
-> addedDate VARCHAR(40)
-> );
Query OK, 0 rows affected (0.00 sec)

```

Output:

The screenshot displays a web browser with two pages. The left page, titled 'Add the item in Inventory', contains a form with the following fields: 'Item Name' (containing 'Laptop'), 'Description' (containing 'HP 15s 8 GB RAM'), and 'Quantity' (containing '10'). An 'AddToInventory' button is at the bottom. The right page, titled 'Click here to add items in the Inventory.', shows a table with the following data:

Item No	Item Name	Description	Quantity	Added Date & Time
3	Laptop	HP 15s 8 GB RAM	20	Sat Aug 11 01:06:35 IST 2018
4	TV	Toshiba LED	30	Sat Aug 11 01:13:24 IST 2018
5	Mobile	Samsung Galaxy Note 2	30	Sat Aug 11 01:14:45 IST 2018
6	Mouse	Logitech Wireless	25	Sat Aug 11 01:20:10 IST 2018
7	Keyboard	Dell USB Keyboard	15	Sat Aug 11 01:25:42 IST 2018

Result:

The simple Inventory Application was successfully developed using **JPA, JSP and MySQL**.